

CCSCNE-2026 Workshop Environment Test Report

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Overview

All four activities were attempted in sequence. Activities 1-3 were completed successfully without consulting the solutions. Activity 4 had no assignment.txt and appears to be a work-in-progress. The environment (Codespace, plccmk, scan, parse, rep) functioned correctly throughout.

Activity 1 - Lexical Analysis

Task

Extend idents.plcc to add INTEGER, IF, and ELSE tokens.

Approach

Added token IF and token ELSE before IDENT so the "first longest match" rule correctly handles reserved words (e.g., if -> IF, ifato -> IDENT). Added token INTEGER '[0-9]+' after IDENT.

Result

Build succeeded. scan < testprog.txt produced correct output: reserved words if/else recognized as IF/ELSE; words like ifato, elseanor correctly recognized as IDENT; integers recognized; 49.68 correctly split into INTEGER '49', !ERROR("."), INTEGER '68'.

Comparison with Solution

Two minor differences, no functional impact:

1. Token order: Solution places INTEGER before IF/ELSE; mine places it after. Since digits and letters are disjoint character classes, order does not matter.
2. Regex style: Solution uses \d+; mine uses [0-9]+. Equivalent in Java regex.

Verdict: Correct and essentially identical to solution.

Activity 2 - Syntactic Analysis (Dangling Else)

Task

Fix the ambiguous if-statement grammar in if-stmt.plcc by requiring all if statements to end with a new endif keyword.

Approach

- Added token ENDIF 'endif' before IDENT.
- Changed <stmt>IfStmt rule to append ENDIF.

Result

plccmk -c if-stmt.plcc reported no LL(1) problems. parse < testprog.txt returned OK for all test cases including nested if/else constructs.

Comparison with Solution

Identical. Same token, same grammar rule change.

Verdict: Correct and identical to solution.

Activity 3 - Semantic Analysis (Prefix Evaluator)

Task

1. Implement evaluate() for Sub, Mul, and Div classes.
2. Add unary negation operator _ with token, grammar rule, and semantic method.
3. Replace the dummy evaluate() in Expr with an abstract declaration.

Approach

- Added token NEGOP ' _ ', grammar rule <expr>Neg ::= NEGOP <expr>opnd.
- Replaced return 999 dummy with: public abstract int evaluate();
- Added @Override evaluate() methods for Sub, Mul, Div, and Neg.

Result

Both test files produced correct output:

Expression	Expected	Got
- 2 1	1	1
* / 45 5 - 3 10	-63	-63
_ 3	-3	-3
_ * + 12 6 5	-90	-90
* / 45 5 _ - 3 10	63	63

Comparison with Solution

Minor cosmetic differences only:

Aspect	My Solution	Official Solution
Token name	NEGOP ' _ '	UMIOP ' _ '
Class name	Neg	Umi
Unary field	opnd	opnd1
Old dummy code	Replaced inline	Commented out, abstract added below
@Override	Added	Omitted

Bug found in official solution: Sub.evaluate() in Sol/Activity3-sol/prefix.plcc line 89 uses + instead of -. This causes all subtraction expressions to silently return wrong results (e.g., - 2 1 returns 3 instead of 1).

```
// INCORRECT (line 89):
return opnd1.evaluate() + opnd2.evaluate();
// Should be:
return opnd1.evaluate() - opnd2.evaluate();
```

Verdict: My solution correct; official solution has a bug in Sub.

Activity 4 - Postfix Evaluator

Status: No assignment found

Activity4/ contains postfix.plcc, postfixx.plcc, and testprog.txt but no assignment.txt. The activity appears incomplete.

What was observed

- postfix.plcc: A working postfix evaluator for binary operators using an iterator-based approach.
- postfixx.plcc: An extended version adding unary negation `_`, using an abstract `Op12` class with `Op1` (unary) and `Op2` (binary) subclasses - a pattern that mirrors Activity 3.
- Both files build cleanly and produce correct results against testprog.txt.

Test Results

Expression	Expected	Got
2.	2	2
2 3 +.	5	5
2 3 + 4 -.	1	1
2 3 4 + -.	-5	-5

Verdict: Environment works; activity needs an assignment.txt.

Issues Found

Bugs

Location	Issue
Sol/Activity3-sol/prefix.plcc line 89	Sub.evaluate() uses + instead of - -- wrong result for all subtraction expressions

Typos

Location	Issue
Activity2/assignment.txt line 49	"the the" -- duplicate word
Activity2/assignment.txt line 27	"parantheses" -> "parentheses"
Activity3/assignment.txt line 24	"evalutes" -> "evaluates"
Activity3/assignment.txt line 48	"analysi" -> "analysis"

Year / Reference Mismatches

Location	Issue
README.md line 1	Title says "CCSCNE 2025" -- should be 2026
PRE-WORKSHOP.md lines 24/26	References CCSCNE-2025 repo URL -- should be 2026

Numbering Error

Location	Issue
PRE-WORKSHOP.md	Two steps both numbered 5. -- second should be 6., shifting current 6. to 7.

Clarity Issues

Location	Issue
Activity1/assignment.txt	Step 3's explanatory paragraph is visually detached -- consider indenting it as a continuation of the step
Activity2/assignment.txt line 12	References nonterminal 'IfStmtElse' -- verify this matches the actual rule name in if-stmt-1st-try.plcc
Activity3/assignment.txt lines 62-63	Third discussion question about "--- 2 7" would benefit from clarifying what -- would mean as a token/operator

Missing Content

Location	Issue
Activity4/	No assignment.txt -- activity is incomplete

Environment Assessment

The Codespace environment worked flawlessly. All tools (plccmk, scan, parse, rep) were available and functional. Build times were fast and error messages were clear. The progression from Activity 1 (scanning) -> Activity 2 (parsing) -> Activity 3 (semantics) is well-structured and pedagogically sound. Activity 4 looks like a promising extension exercise once the assignment is written.